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Motivation

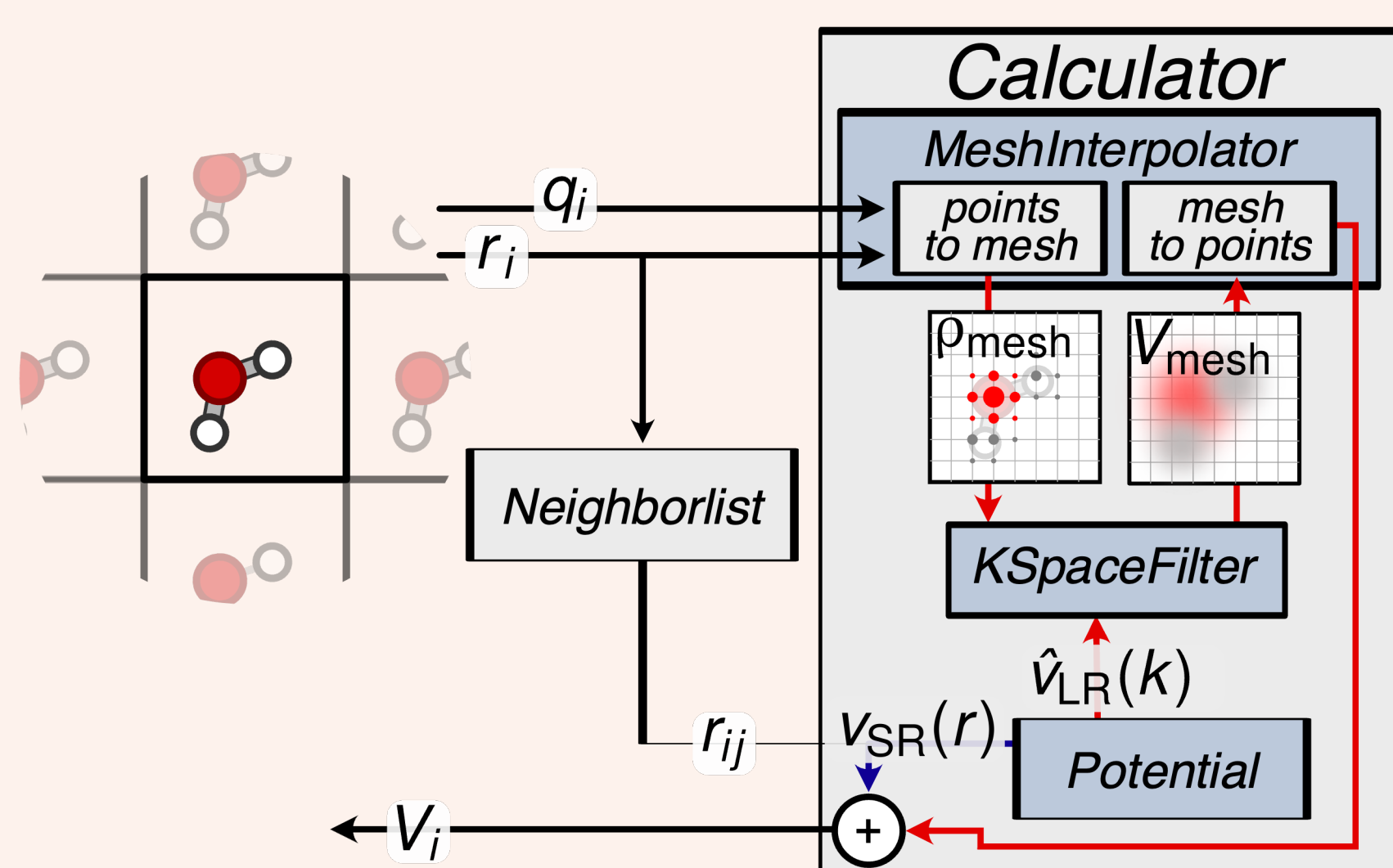
- ⚡ ML surrogate models can eliminate the need for costly quantum mechanical simulations.
- 🌀 Atomistic ML models typically rely on local neighborhoods (within a specified cutoff) to predict atomic properties.
- 👉 To account for long-range interactions, a modular framework is needed to integrate these effects into existing architectures.
- 🌱 Incorporating such a framework enhances the accuracy of current models without requiring extensive modifications.

Modular software for custom ML workflows



tab-cosmo/**torch-pme**

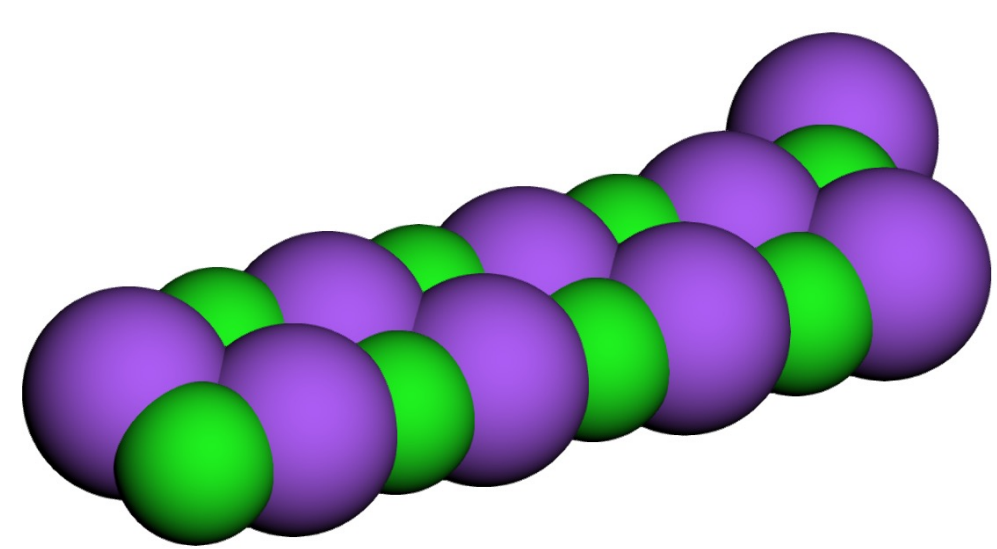
- Plug-and-play Python module for long-range interactions
- Efficient, auto-differentiable, optimized for CPU & GPU
- Implementations for direct all-vs-all summation, Ewald summation, PME, and P3M



metatensor/**metatrain**

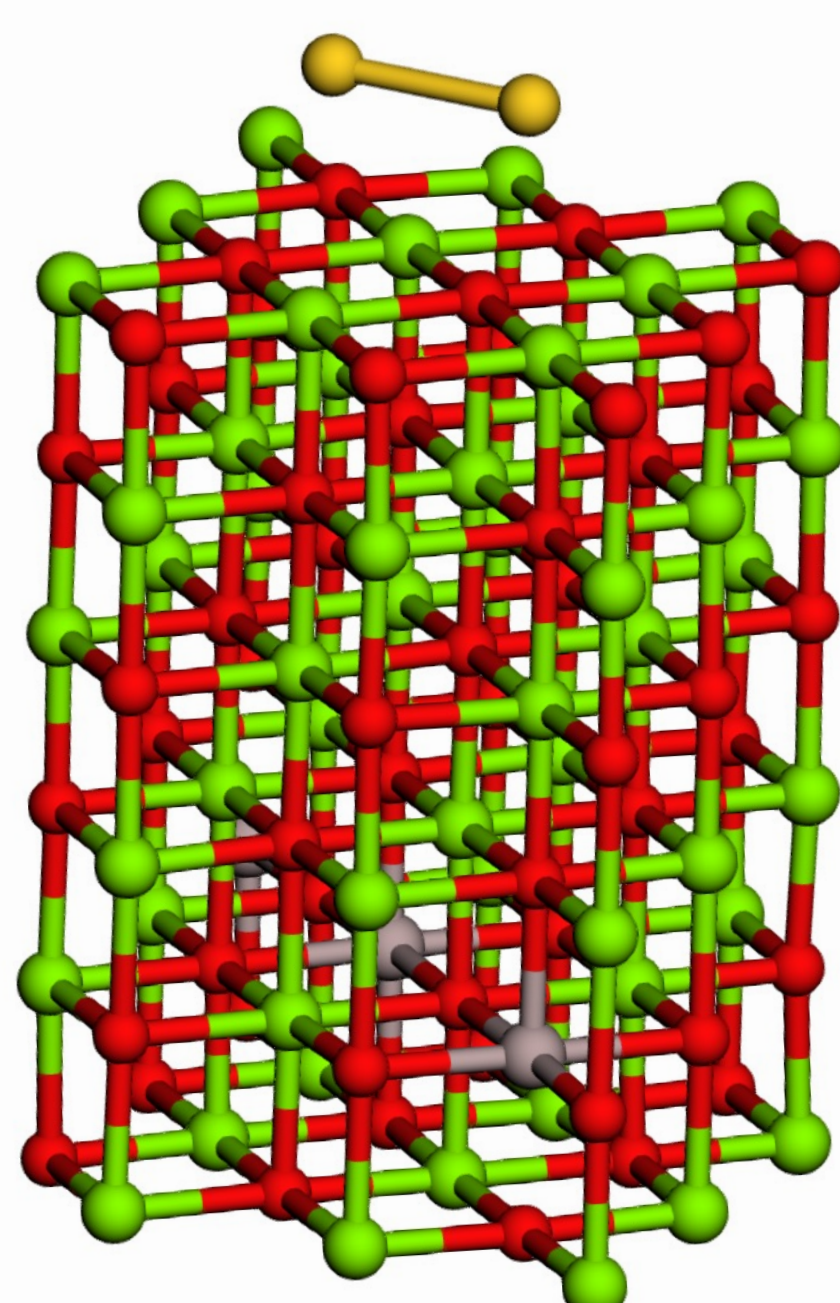
- User-friendly command-line interface for training and evaluating atomistic models
- Modular and intuitive structure
- Supports advanced functionalities, including fitting to general target properties
- Integrates **torch-pme**, enabling efficient long-range interaction calculations for SOAP-BPNN and nanoPET models

Datasets used for evaluation

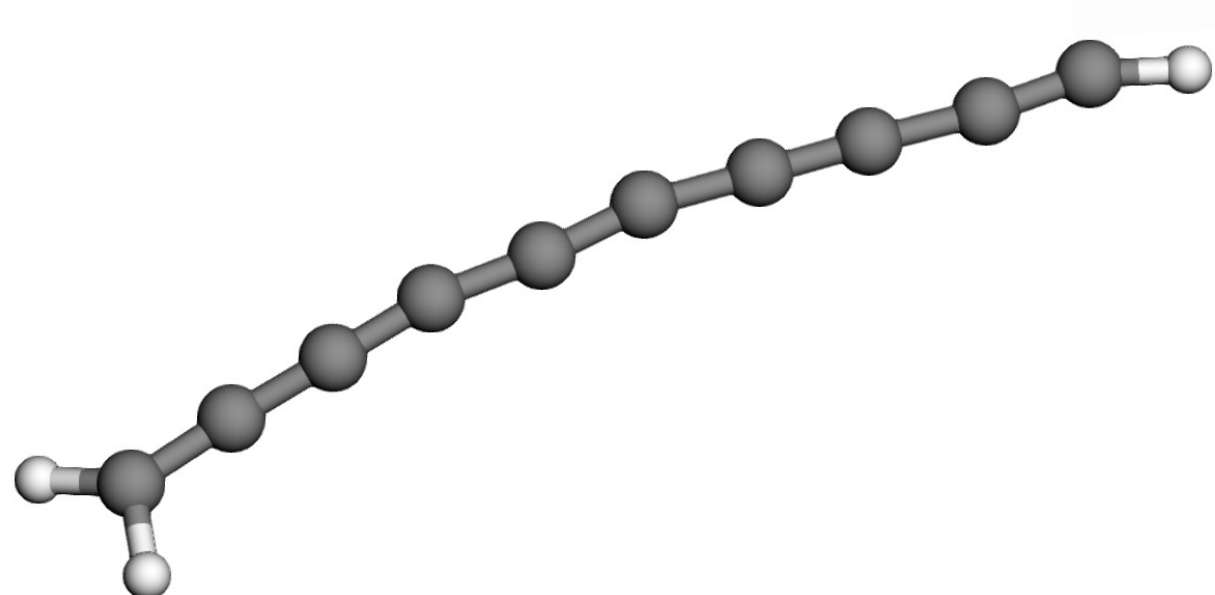


NaCl clusters

Au dimer on (un)doped MgO crystal



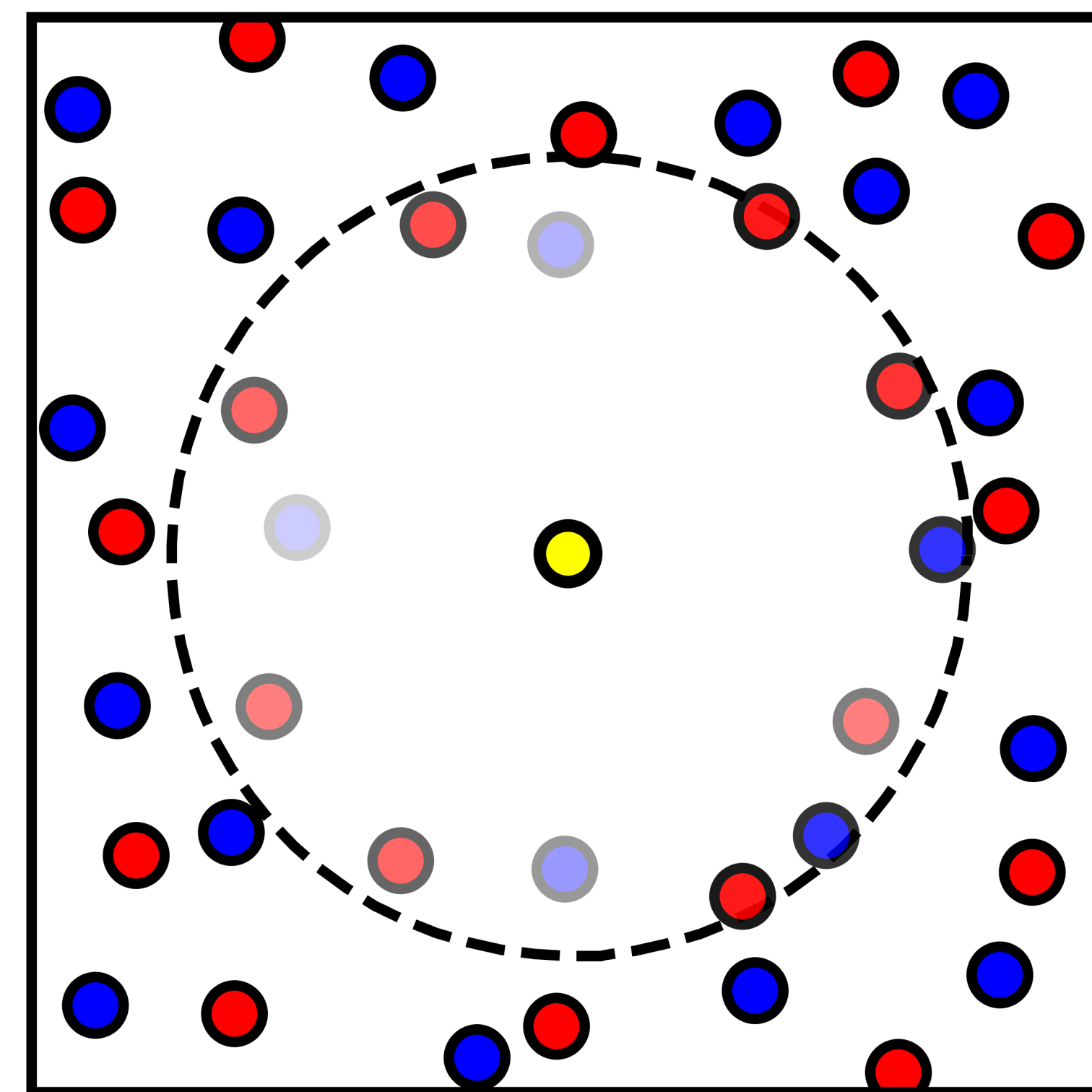
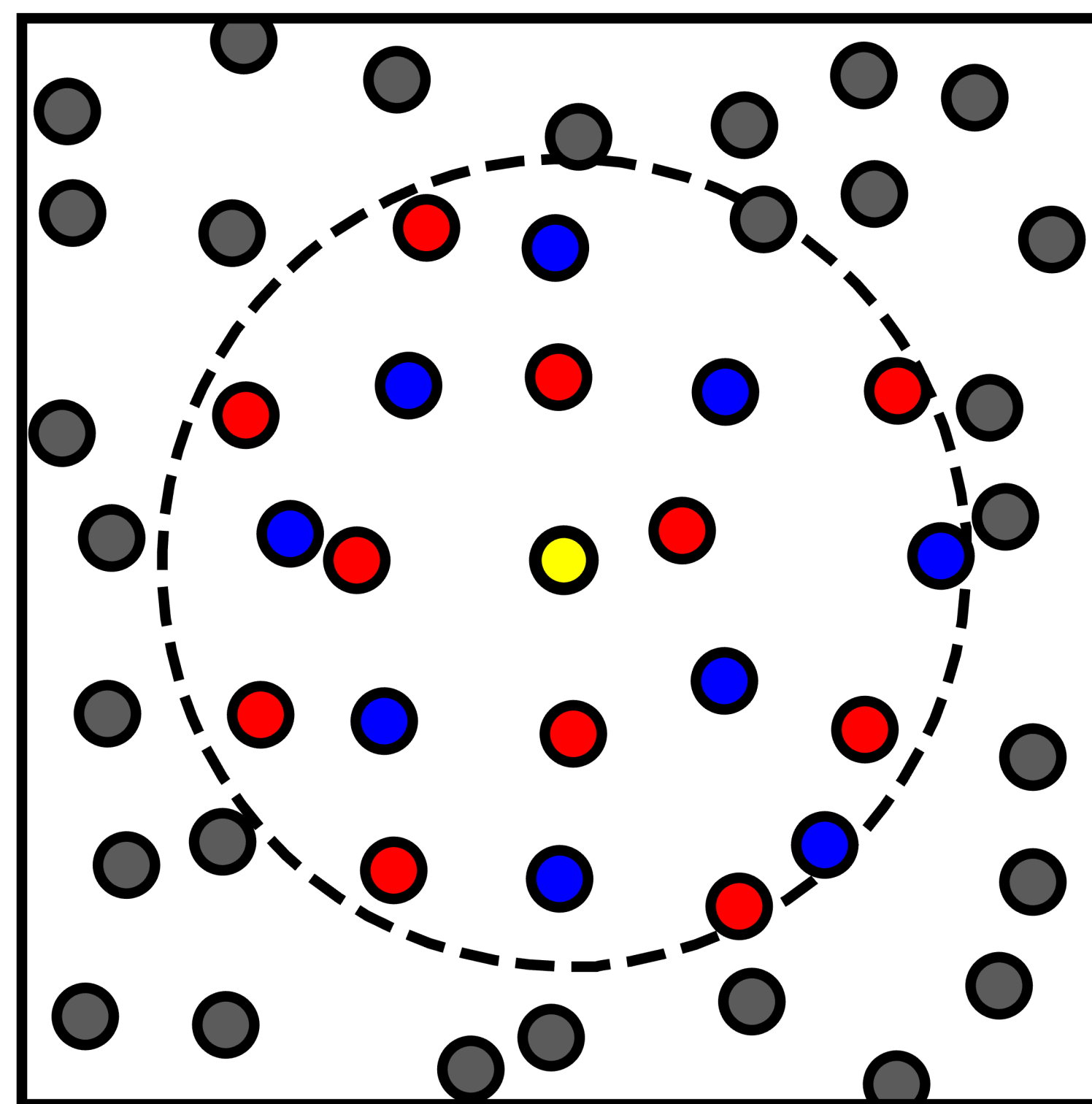
Carbon chain



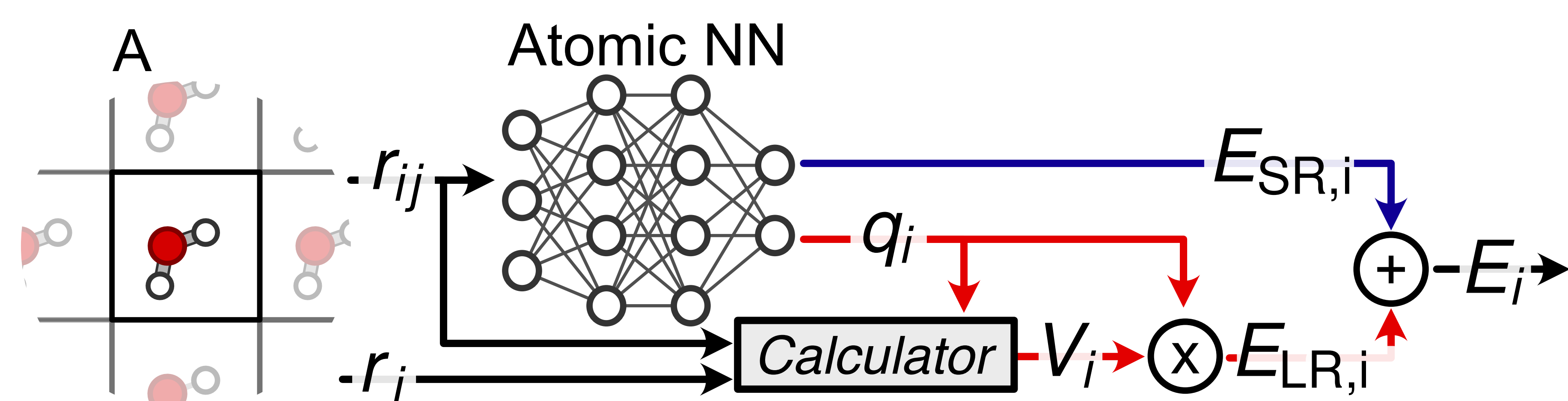
Nearsightedness of machine-learned interatomic potentials

Local atomic environment for property prediction

Extended atomic environment captured by **torch-pme**



Integration of long-range features into existing models

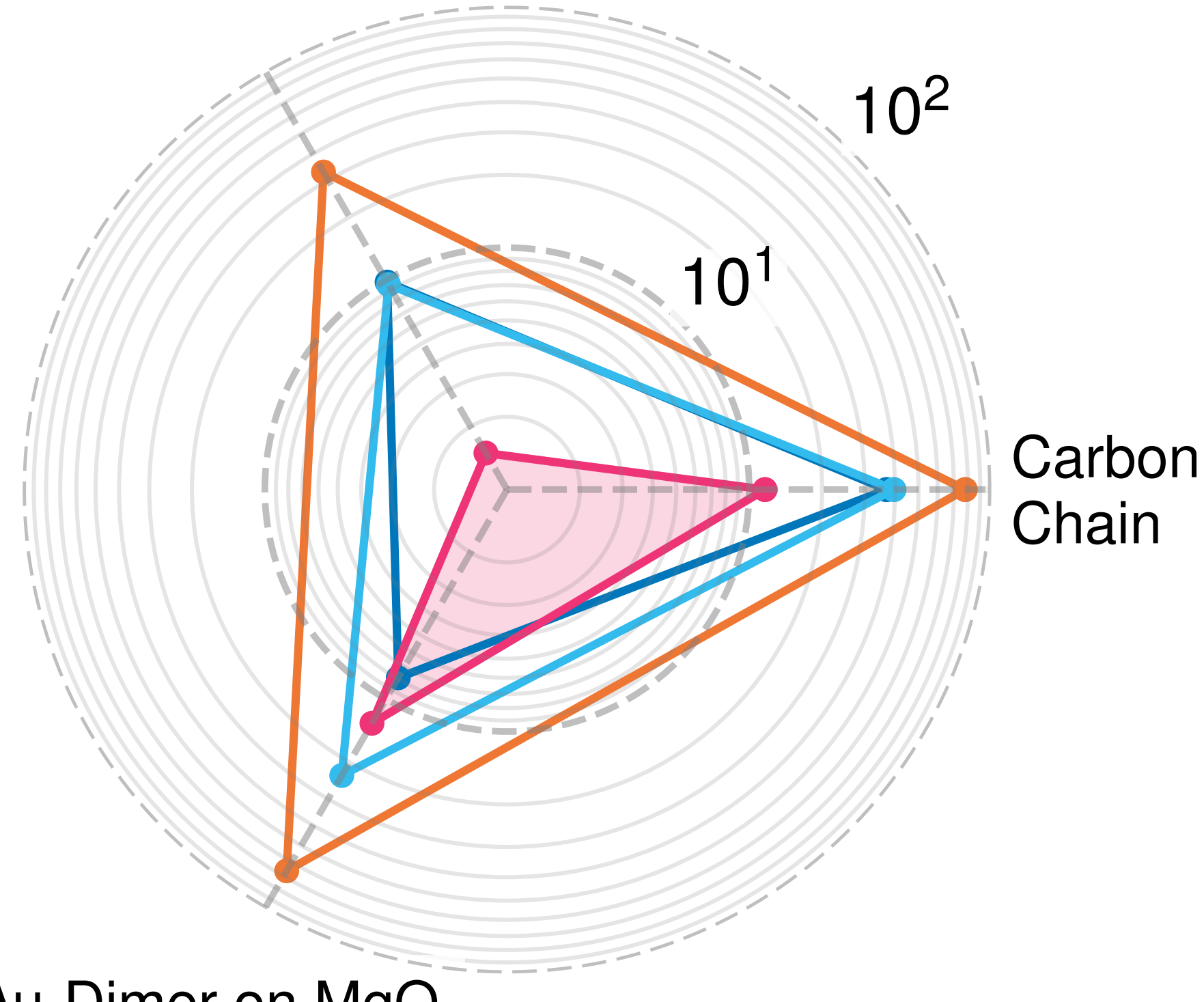
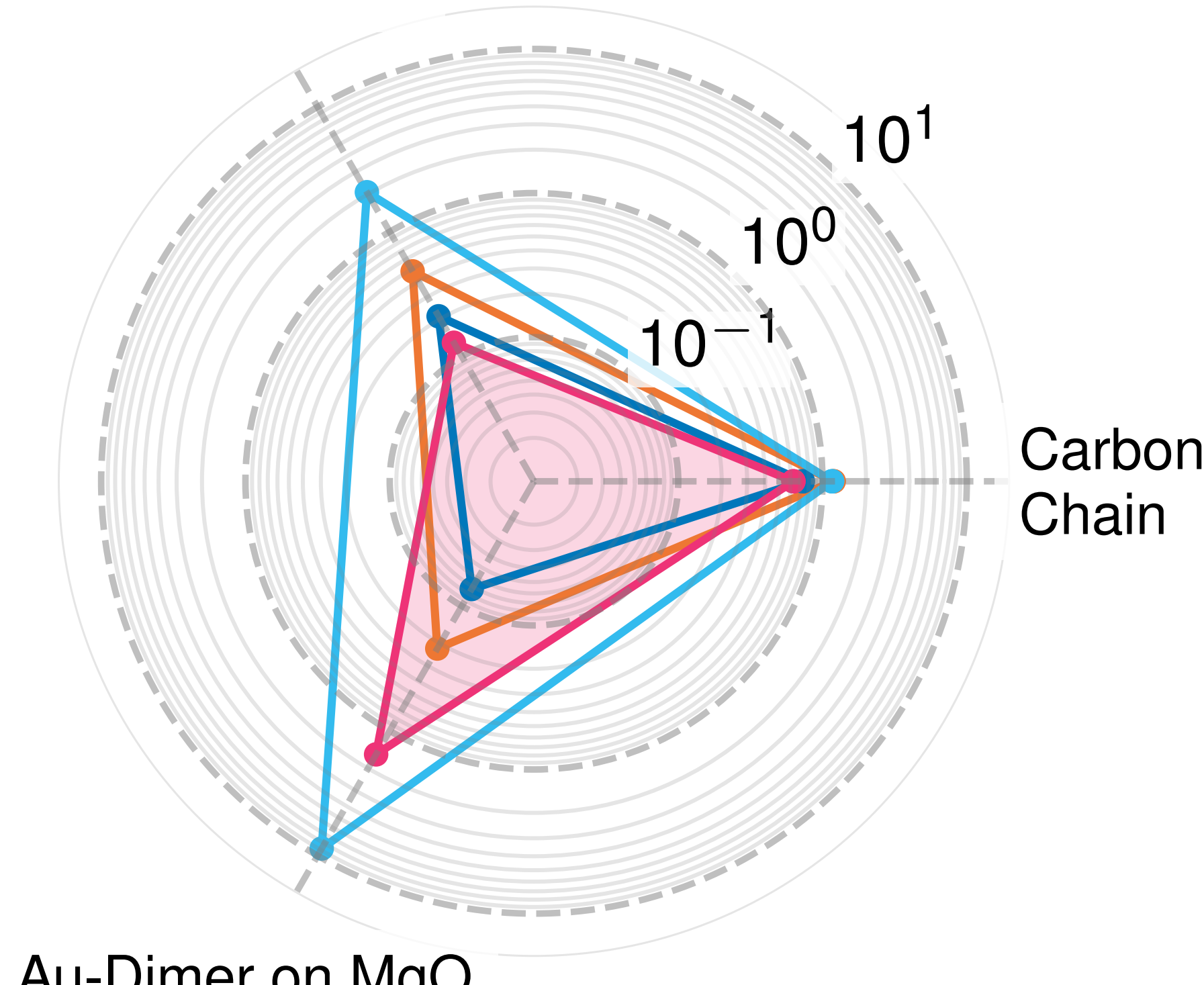


Energy Error [meV/atom]

Force Error [meV/Å]

NaCl Clusters

NaCl Clusters

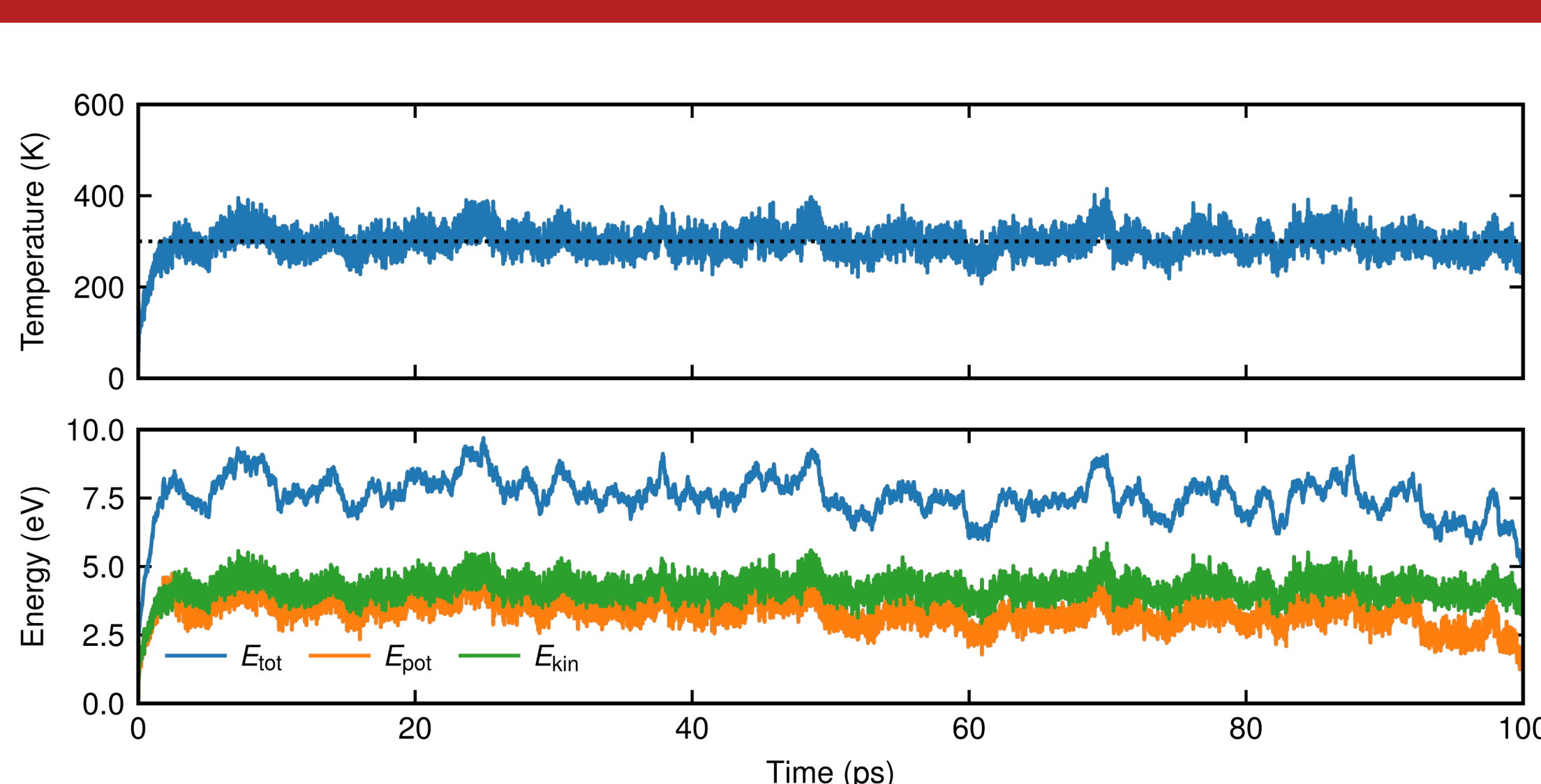


Au-Dimer on MgO

Au-Dimer on MgO

— 4G-HDNNP — CACE-LR — SOAP-LR — nanoPET-LR

Test MD run on Au dimer on MgO



What's next?

1. Investigate various architectural modifications to enhance the performance of long-range models.
2. Develop robust and transferable long-range model training protocol.
3. Curate extensive, high-quality datasets designed to fully exploit the capabilities of models incorporating long-range interactions.
4. Develop universal long-range machine-learned interatomic potentials